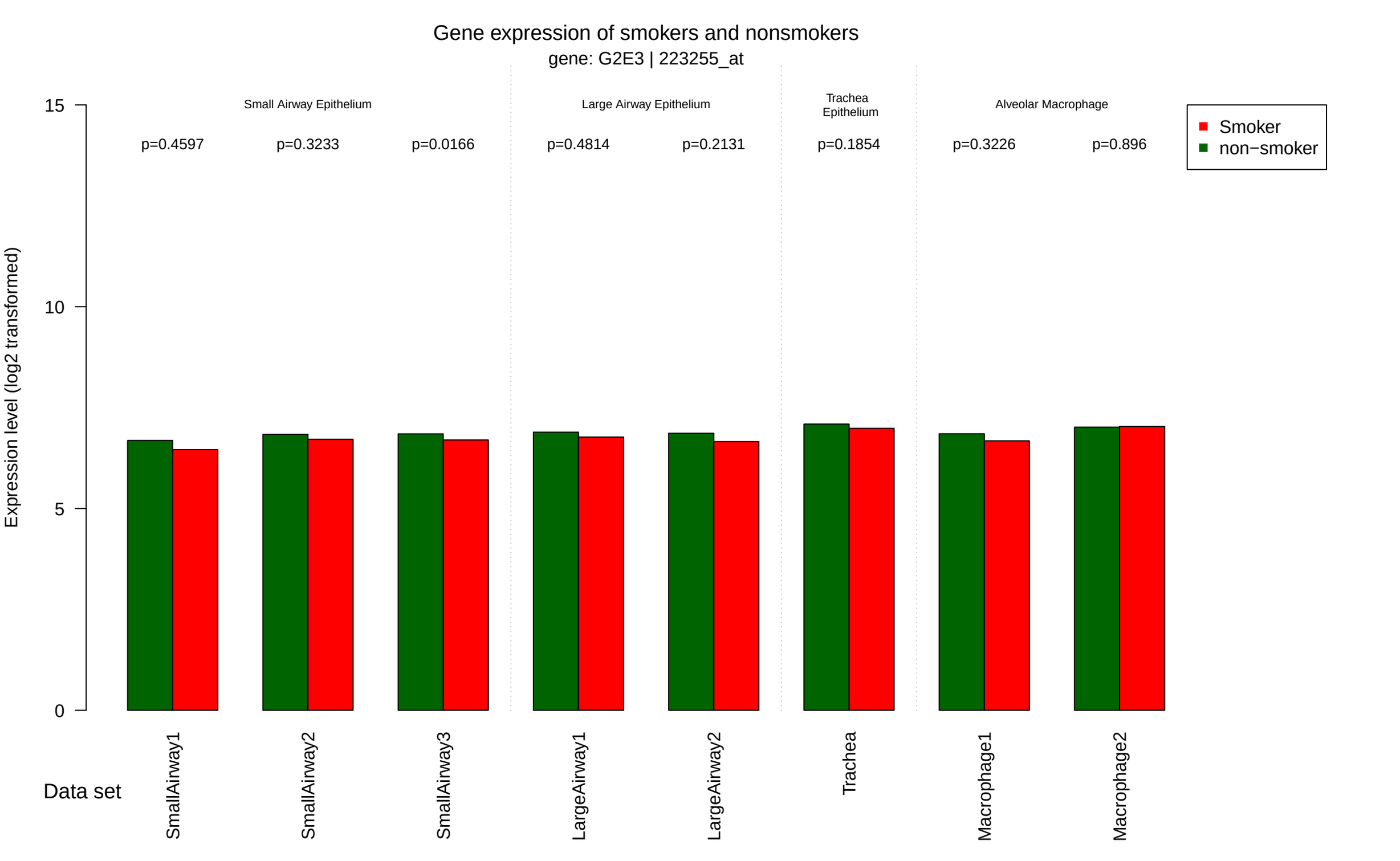
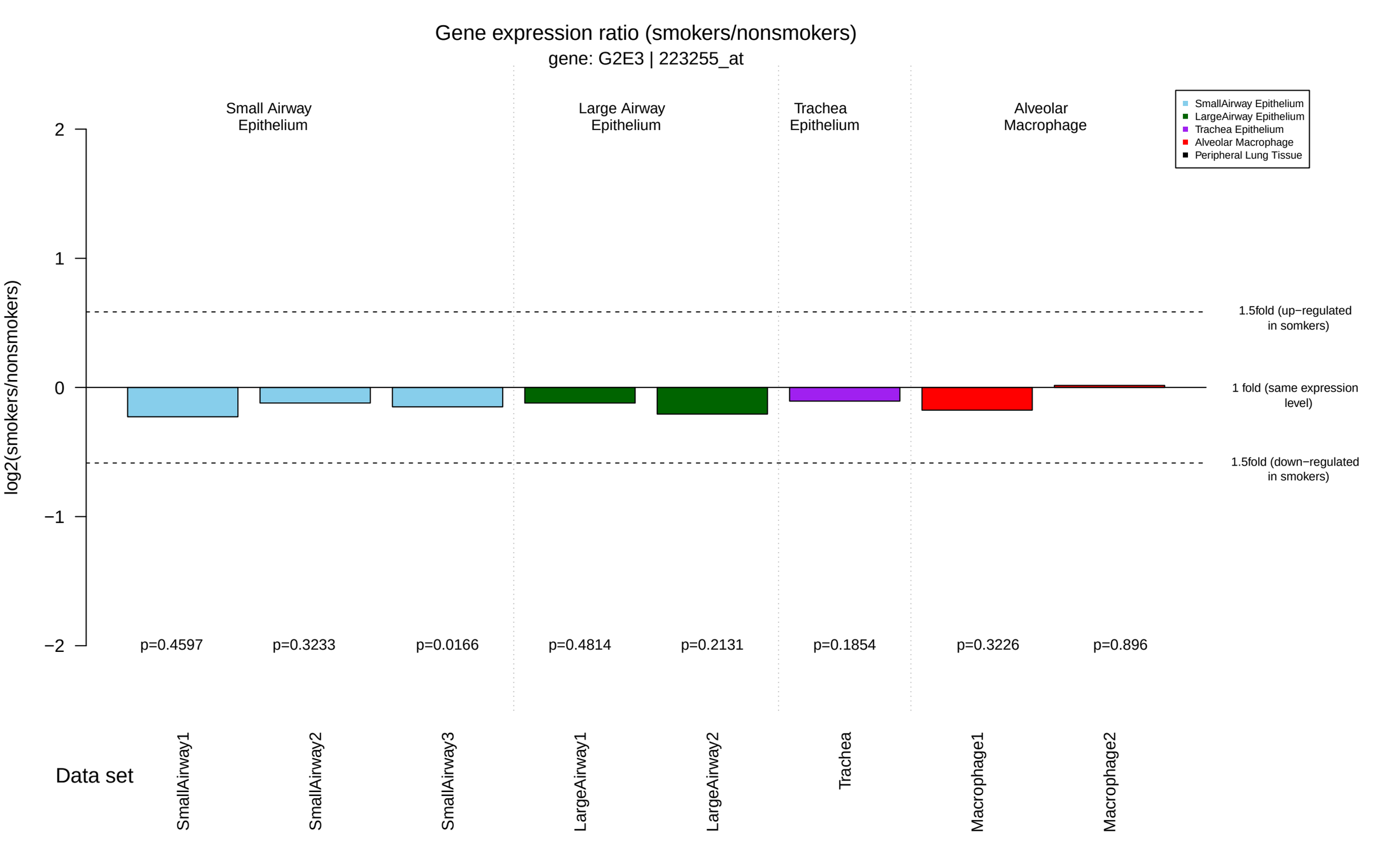
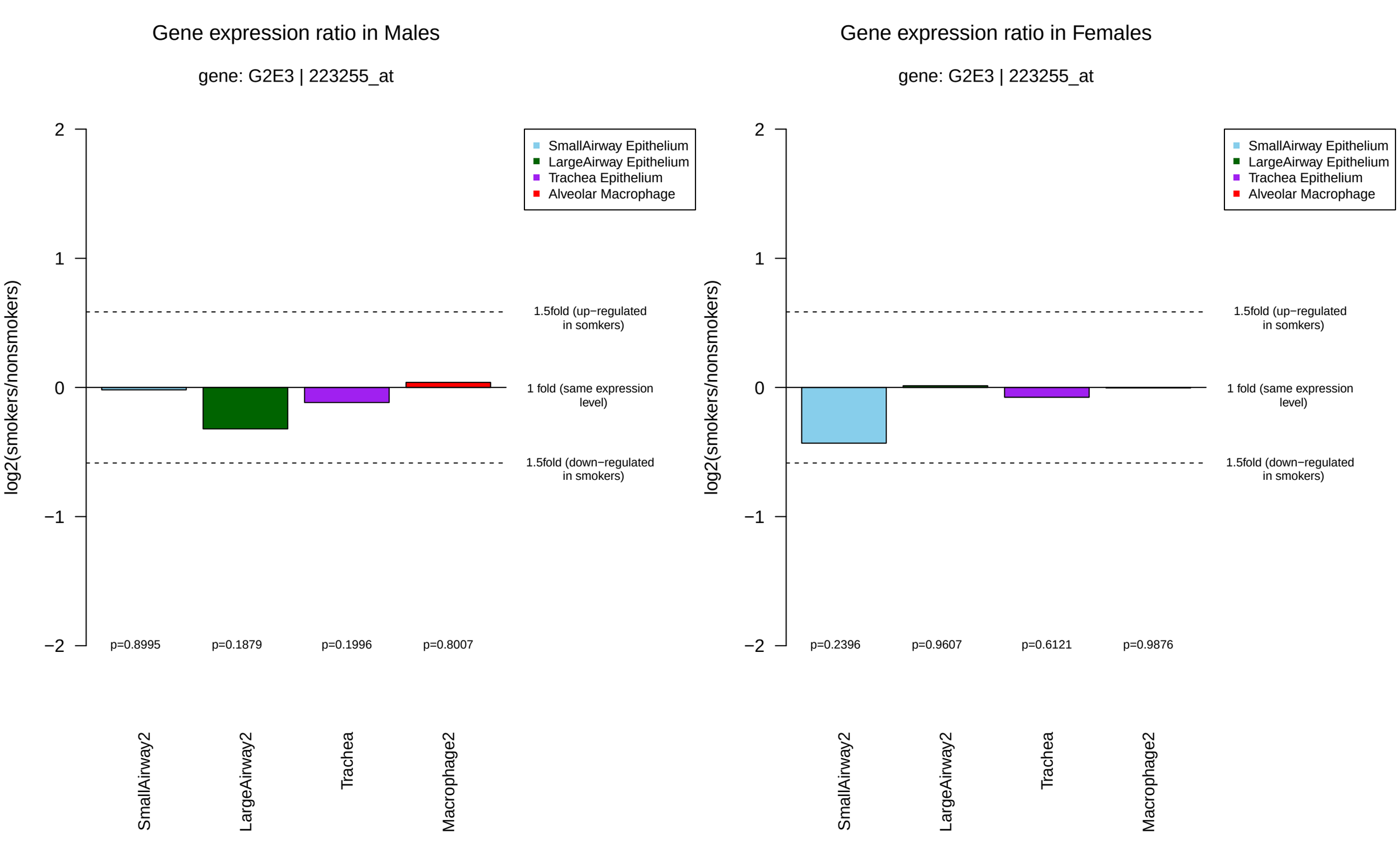
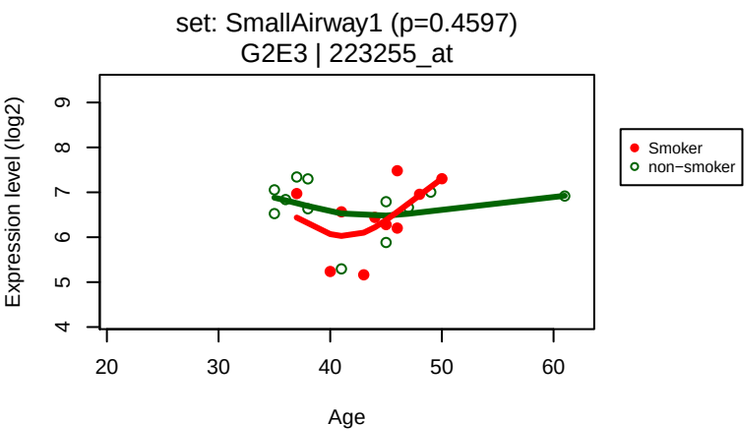
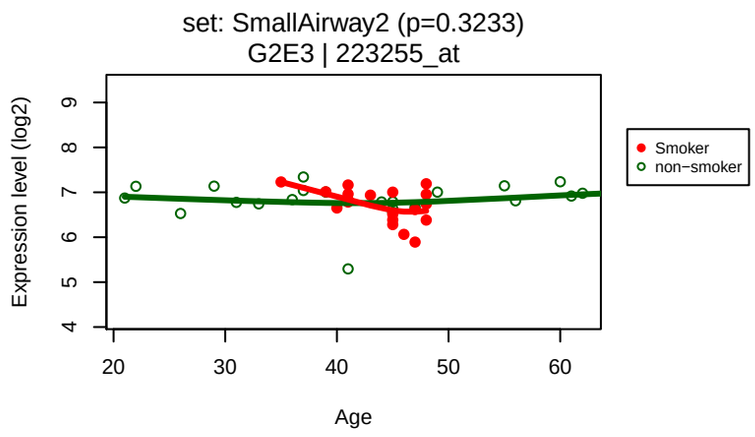
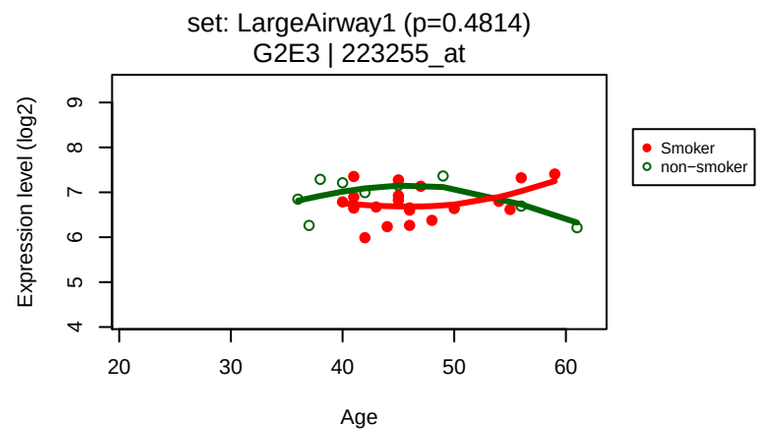

Gene expression of smokers and nonsmokers
gene: G2E3 | 223255_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4597
p=0.3233
p=0.0166
p=0.4814
p=0.2131
p=0.1854
p=0.3226
p=0.896
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

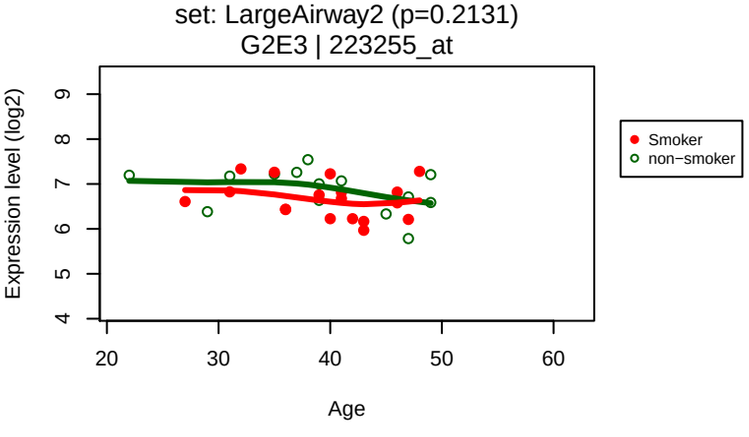

Gene expression ratio (smokers/nonsmokers)
gene: G2E3 | 223255_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4597
p=0.3233
p=0.0166
p=0.4814
p=0.2131
p=0.1854
p=0.3226
p=0.896
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

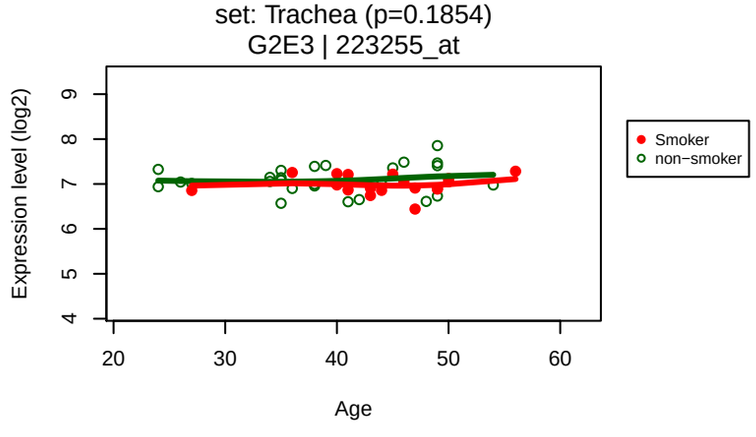

Gene expression ratio in Males
gene: G2E3 | 223255_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8995
p=0.1879
p=0.1996
p=0.8007
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: G2E3 | 223255_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2396
p=0.9607
p=0.6121
p=0.9876
SmallAirway2
LargeAirway2
Trachea
Macrophage2

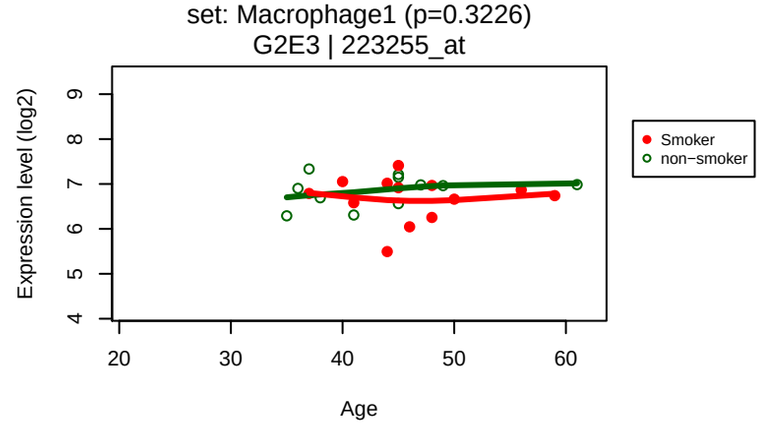

set: SmallAirway1 (p=0.4597)
G2E3 | 223255_at
Expression level (log2)
Age
Smoker
non−smoker

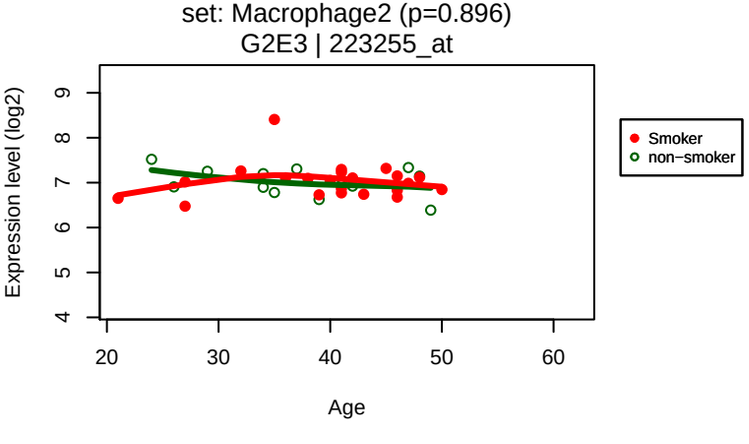

set: SmallAirway2 (p=0.3233)
G2E3 | 223255_at
Expression level (log2)
Age
Smoker
non−smoker

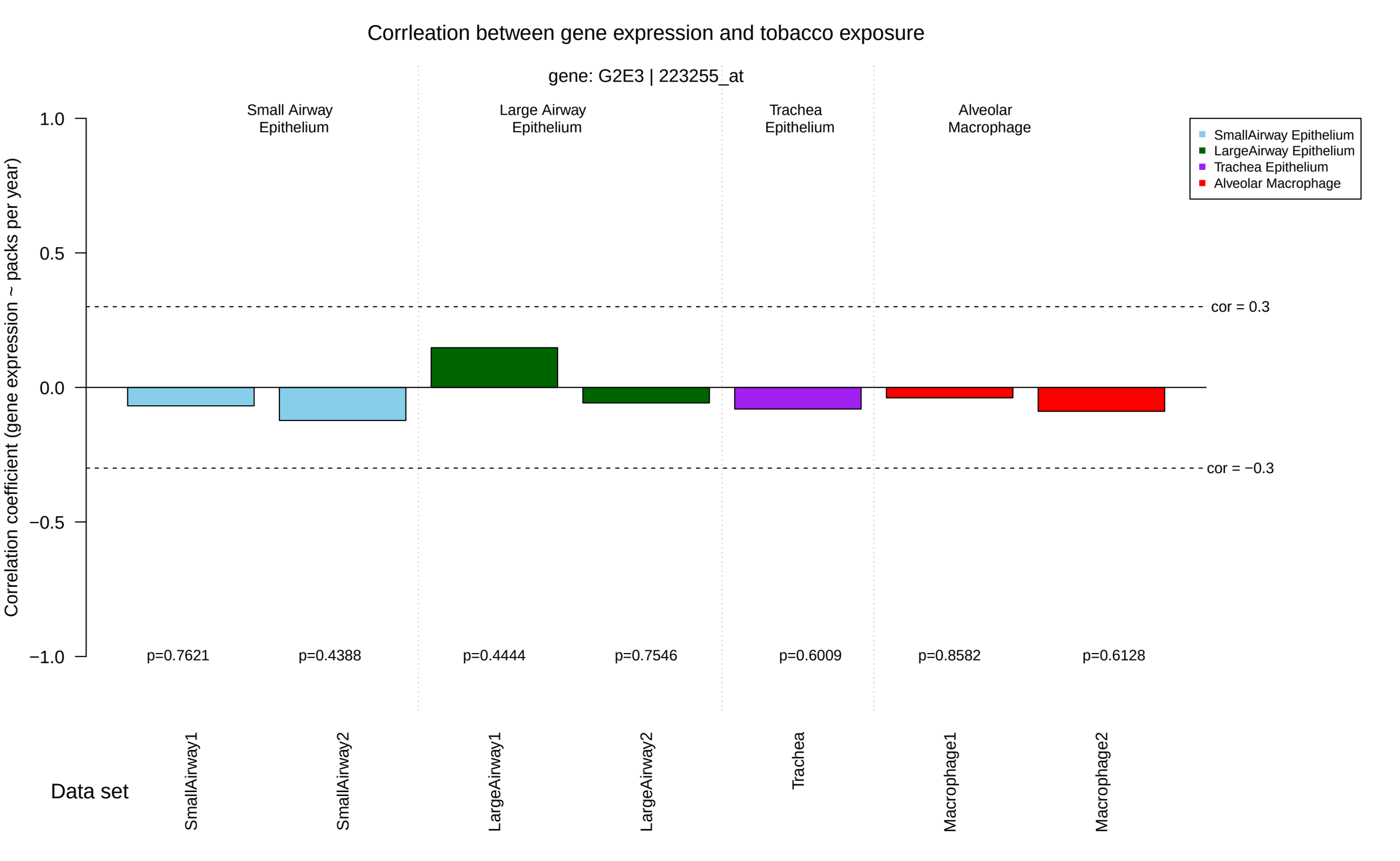

set: LargeAirway1 (p=0.4814)
G2E3 | 223255_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2131)
G2E3 | 223255_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1854)
G2E3 | 223255_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3226)
G2E3 | 223255_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.896)
G2E3 | 223255_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: G2E3 | 223255_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7621
p=0.4388
p=0.4444
p=0.7546
p=0.6009
p=0.8582
p=0.6128
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2